Calling Data Integration: Problem, Solution and Roadmap

Oct 30, 2026 · @Ankit lal

Problem statement

Our calling operation generates call data we cannot get out. Agents dial through X-Lite (a CounterPath softphone) into a CASQ 32-port GSM / VoLTE VoIP gateway, and none of that activity reaches our own reporting panel.

Current setup

- **Software:** X-Lite softphone by CounterPath, installed on each agent machine.
- **Hardware:** CASQ multi-port GSM / VoLTE VoIP gateway, 32 ports / 32 channels (SWG-3032 series), one SIM per port.
- **Reporting:** none centralised. Each softphone keeps only its own local call history.

What is broken

- Call data (who dialled, which number, duration, outcome) is locked inside individual X-Lite installs.
- CounterPath has been unsupportive and offers no route to export or integrate this data. X-Lite is also a discontinued product, so no fix is coming.
- We cannot track agent productivity, connect rates, talk time or SIM usage in one place.
- Campaign decisions are made without performance data.

What we need

An open, self-hosted calling layer that works with our existing CASQ gateway and stores every call in a database our team controls. Our developers must be able to build on it and feed an external reporting dashboard.

Proposed solution

Replace X-Lite with an open-source dialer server that sits between our agents and the CASQ gateway. Every call then passes through a system we own, and every call record lands in our own database.

The gateway stays as it is. It is a standard SIP device, so it connects to any SIP platform as a trunk. Only the software layer changes.

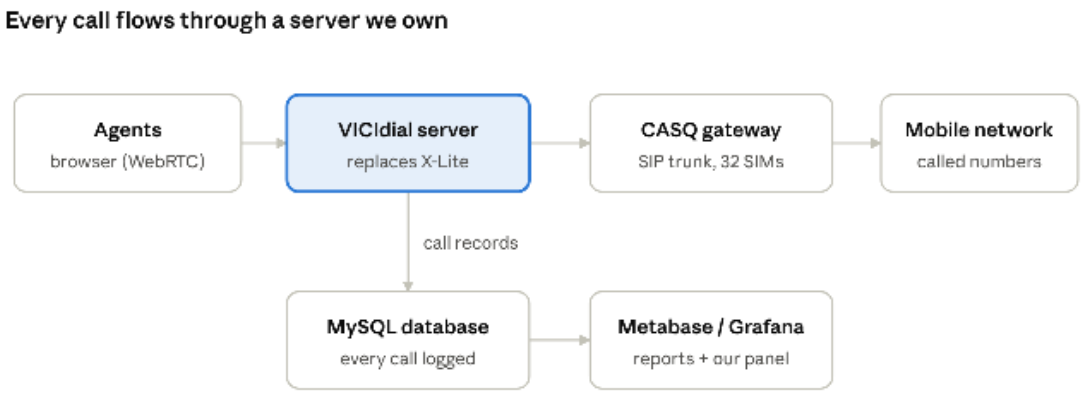
Recommended stack

Layer	Recommended	Alternative	Why

Dialer / PBX	VICIdial	Asterisk + FreePBX	Built for outbound campaigns: lead lists, predictive dialling, dispositions, recordings, reports
Agent phone	VICIdial web agent (WebRTC)	MicroSIP or Linphone	Free, no per-seat licence, no vendor lock-in
Database	MySQL (ships with VICIdial)	PostgreSQL	Every call, agent session and disposition stored as rows
Dashboard	Metabase or Grafana	Custom panel via API	Open source, connects straight to MySQL
Integration	VICIdial agent and non-agent APIs	Asterisk AMI / ARI	Real-time events and data pulls for our own panel

Why VICIdial over plain FreePBX: our use case is a calling operation, not an office phone system. VICIdial already handles campaigns, lists, agent login and call outcomes, so our developers build reporting rather than rebuilding a dialer.

How the pieces connect:



target architecture · gateway unchanged, software layer replaced

The CASQ gateway stays in place; VICIdial takes over from X-Lite and writes every call to a database our dashboard reads.

Stepwise process

We move in six phases and keep X-Lite running until the new system is proven on live calls.

1. **Audit the gateway and recover existing data**
    - Log into the CASQ web panel and note firmware version, SIP settings and port status.

- Check the CDR section. Gateways in this class store about 10,000 call records locally and export them as CSV. Download this now as our baseline.
- Note per-port statistics (ASR, ACD) to spot weak SIMs.

1. **Set up the server**
    - Provision a Linux server on the same office network as the gateway (a mid-range machine or local VM is enough for 32 channels).
    - Install VICIdial using the ViciBox ISO or a scratch install on a supported distro.
    - Fix a static IP for both the server and the gateway.
1. **Connect the gateway as a SIP trunk**
    - On the CASQ panel, point the SIP trunk at the VICIdial server.
    - On VICIdial, add a carrier entry for the gateway with matching codec (G.711 / G.729) and DTMF settings.
    - Place test calls through a single port, then through all 32.
1. **Configure campaigns and agents**
    - Create 14 user accounts (10 callers, 2 managers, 2 admins), agent phones and at least one campaign.
    - Upload a test lead list and define dispositions (connected, no answer, busy, callback, wrong number).
    - Agents log in through the browser; no softphone install needed.
1. **Pilot alongside X-Lite**
    - Move a few agents to VICIdial for one week while the rest stay on X-Lite.
    - Compare call quality, connect rates and agent feedback.
    - Confirm every call shows up correctly in the VICIdial database.
1. **Build the reporting dashboard and cut over**
    - Install Metabase or Grafana and connect it read-only to the MySQL database.
    - Build core views: calls per agent, talk time, connect rate, outcomes by campaign, usage per SIM / port.
    - For our own panel, developers pull data through the VICIdial APIs or scheduled queries.
    - Move all agents over and retire X-Lite.

Team needed: one person with Linux and VoIP (Asterisk) experience for phases 2 to 4, and one developer for phase 6.

Risks and next steps

Risk	Mitigation
No in-house VoIP skills	Hire a VICIdial freelancer for setup; many work remotely in India
Downtime during switch	Run the pilot in parallel; keep X-Lite live until cut-over
Call recordings and lead data are sensitive	Restrict server access, back up the database daily

Immediate actions

- Export the existing CDR CSV from the CASQ panel
- Confirm VICIdial as the platform (lead lists and repeat calling rule out plain FreePBX)
- Arrange a server and assign a VoIP person
- Fix pilot agents and pilot dates

Team and access

14 accounts in total. With 10 callers the 32-port gateway has ample spare capacity, so a modest server is enough.

Role	Accounts	Access
Callers	10	Agent screen only: log in, dial, set call outcomes
Managers	2	Live monitoring, campaign and lead list control, all reports
Admins	2	Full system access: users, carrier / gateway settings, server

Dialling approach and repeat contacts

We start with manual dialling and move to pre-uploaded lead lists later. The same person will be called several times over a period, so the system must track each person across every call and every campaign.

Stage	How callers dial	What VICIdial does
Short term	Manual: caller types or picks the number	Logs every call with caller, time, duration and outcome
Long term	Lead lists uploaded by managers	Pushes the next due contact to a free caller; schedules follow-ups automatically

Designing for repeat calls

- **One contact, one record.** The phone number is the unique key. A person uploaded twice

is merged, not duplicated; VICIdial's duplicate checks can enforce this across lists.

- **Full call history per person.** Every call is logged against that contact, so a caller sees past outcomes and notes before dialling.
- **Rounds, not one-off campaigns.** Each calling round (first contact, follow-up, reminder) is a separate campaign or list working off the same contact base.
- **Scheduled follow-ups.** Callers set callbacks for a date; managers recycle leads by outcome (for example, retry no-answers after two days).
- **Contact-level reporting.** The dashboard shows each person's journey: attempts, rounds reached, latest outcome and gaps since last contact, alongside the usual caller and campaign views.

Build note for developers: treat the contact as the core entity in our panel and join all call logs to it by phone number. This keeps history intact even when lists or campaigns change.